From DNA to Protein

Biology 2e chapter 15 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

Cell-free reactions contain molecular machinery outside intact cells. A fluorescent reporter can make a product detectable. A negative control tests background signal; fluorescence also depends on folding and maturation, not just transcription.

Investigation question

What evidence links a DNA template and translation to production of a fluorescent protein?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Observe the same reaction volume and incubation conditions for all tubes. Switch the viewer on to inspect fluorescence. A generic teaching model is used; the chosen classroom kit determines actual timing and reagents.

Prepare the equipment in the displayed order. Observe or classify the sample, review your measurements, then click Collect data and continue. Repeat through the assigned comparison. Explore other settings after collecting the full comparison.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: relative fluorescence

Condition / sample	Repeat	Source	relative fluorescence
Reporter DNA	1		
Reporter DNA	2		
Reporter DNA	3		
No DNA control	1		
No DNA control	2		
No DNA control	3		
Reporter DNA + translation inhibitor	1		
Reporter DNA + translation inhibitor	2		
Reporter DNA + translation inhibitor	3		

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Copy this page for each trial. Treatment: _____ Repeat: ____ Source: _____

hours	relative fluorescence	Notes
0		
2		
4		
6		
8		

Graph your observations

Label axes and units; keep each treatment and source identifiable. For repeated measurements, connect readings only from the same sample. For ordinal scores, explain the categories instead of treating the scale as concentration.

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What does the no-DNA control tell you about background fluorescence?

How does the inhibitor comparison strengthen the interpretation?

Why does fluorescence alone not directly measure an mRNA concentration?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Generic single-reporter model in arbitrary units; not a reproduction or calibration of BioBits. Transcription and translation are not independently measured. Classroom ordinal brightness and numerical virtual fluorescence must not be pooled.